How to calculate precision and recall in a 3 x 3 confusion matrix

Asked 7 years ago Active 1 year, 3 months ago Viewed 41k times

15

6

```
                     Predicted
                       class
                  Cat  Dog Rabbit
  Actual class
              Cat   5    3      0
              Dog   2    3      1
           Rabbit   0    2     11
```

How can I calculate precision and recall so It become easy to calculate F1-score. The normal confusion matrix is a 2 x 2 dimension. However, when it become 3 x 3 I don't know how to calculate precision and recall.

machine-learning precision-recall

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edited Mar 23 '14 at 11:58
TooTone
3,511 23 33

asked Mar 23 '14 at 8:26
user22149
153 1 1 4

5 Answers

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21

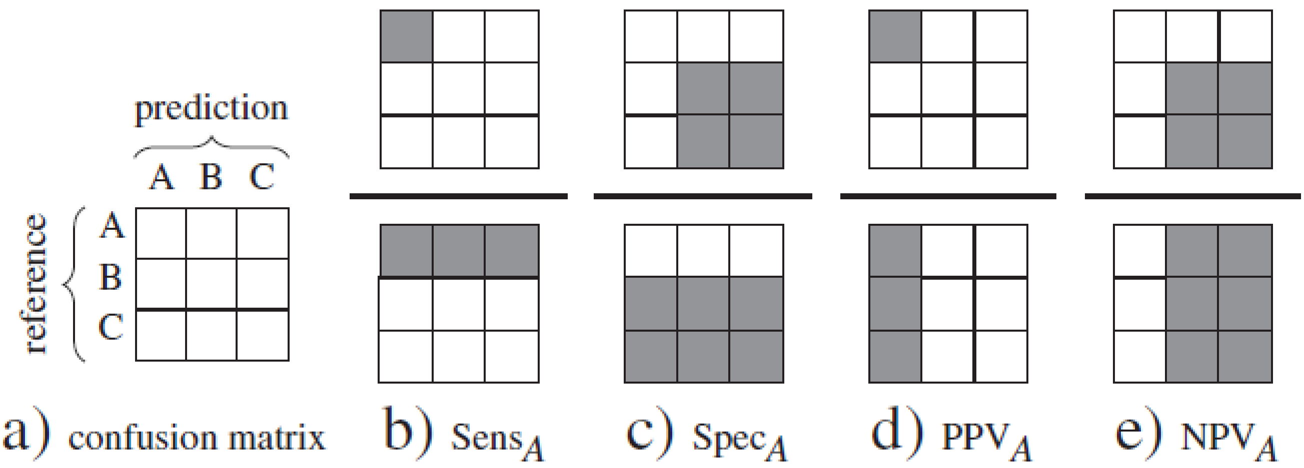
If you spell out the definitions of precision (aka positive predictive value PPV) and recall (aka sensitivity), you see that they relate to *one* class independent of any other classes:

Recall or senstitivity is the proportion of cases correctly identified as belonging to class *c* among all cases that truly belong to class *c*.
(Given we have a case truly belonging to "*c*", what is the probability of predicting this correctly?)

Precision or positive predictive value PPV is the proportion of cases correctly identified as belonging to class *c* among all cases of which the classifier claims that they belong to class *c*.
In other words, of those cases *predicted* to belong to class *c*, which fraction truly belongs to class *c*? (Given the predicion "*c*", what is the probability of being correct?)

negative predictive value NPV of those cases predicted *not* to belong to class *c*, which fraction truly doesn't belong to class *c*? (Given the predicion "not *c*", what is the probability of being correct?)

So you can calculate precision and recall for each of your classes. For multi-class confusion tables, that's the diagonal elements divided by their row and column sums, respectively:



Source: Beleites, C.; Salzer, R. & Sergo, V. Validation of soft classification models using partial class memberships: An extended concept of sensitivity & co. applied to grading of astrocytoma tissues, Chemom Intell Lab Syst, 122, 12 - 22 (2013). DOI: 10.1016/j.chemolab.2012.12.003

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edited Aug 16 '18 at 7:51

answered Mar 23 '14 at 12:11

cbeleites unhappy with SX

32.2k 2 64 125

Thanks so much. I already understand the analogy described in your solution. I will read paper. I will accept this as a answer. I don't understand PPV AND NPV.Please explain these concept as graphic as the Sens and Spec were explained and I will accept your answer. – user22149 Mar 23 '14 at 22:27

3

By reducing the data down to forced choices (classification) and not recording whether any were "close calls", you obtain minimum-information minimum-precision statistical estimates, in addition to secretly assuming a strange utility/loss/cost function and using arbitrary thresholds. It would be far better to use maximum information, which would include the probabilities of class membership and not forced choices.

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answered Mar 23 '14 at 12:24

Frank Harrell

65.7k 4 132 287

1

Following is an example of a multi-class confusion matrix assuming our class labels are A, B and C

A/P	A	B	C	Sum
A	10	3	4	17
B	2	12	6	20
C	6	3	9	18
Sum	18	18	19	55

Now we calculate three values for Precision and Recall each and call them Pa, Pb and Pc; and similarly Ra, Rb, Rc.

We know Precision = TP/(TP+FP), so for Pa true positive will be Actual A predicted as A, i.e., 10, rest of the two cells in that column, whether it is B or C, make False Positive. So

Pa = 10/18 = 0.55 Ra = 10/17 = 0.59

Now precision and recall for class B are Pb and Rb. For class B, true positive is actual B predicted as B, that is the cell containing the value 12 and rest of the two cells in that column make False Positive, so

Pb = 12/18 = 0.67 Rb = 12/20 = 0.6

Similarly Pc = 9/19 = 0.47 Rc = 9/18 = 0.5

The overall performance of the classifier will be determined by average Precision and Average Recall. For this we multiply precision value for each class with the actual number of instances for that class, then add them and divide them with total number of instances. Like ,

Avg Precision = (0.55* 17 + 0.67 * 20 + 0.47 * 18)/55 = 31.21/55 = 0.57 Avg Recall = (0.59* 17 + 0.6 * 20 + 0.5 * 18)/55 = 31.03/55 = 0.56

I hope it helps

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edited Dec 30 '19 at 6:53

answered Dec 29 '19 at 7:30

N

Nayyer Masood

11 2

1

Want to improve this post? Provide detailed answers to this question, including citations and an explanation of why your answer is correct. Answers without enough detail may be edited or deleted.

The easiest way is to not use confusion_matrix at all, Use classification_report(), it will give you everything you ever needed, cheers...

Edit:
this is the format for confusion_matrix():
[[TP,FN]
[FP,TN]]
And classification report gives all this

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edited Jul 27 '18 at 16:30

answered Jul 27 '18 at 12:22

omkaartg

117 3

That if you are using python/// – omkaartg Jul 27 '18 at 12:23

0

If you simply want the result, my advice would be to not think too much about and use the tools at your disposal. Here is how you can do it in Python;

```
import pandas as pd
from sklearn.metrics import classification_report

results = pd.DataFrame(
```

```
    [[1, 1],
     [1, 2],
     [1, 3],
     [2, 1],
     [2, 2],
     [2, 3],
     [3, 1],
     [3, 2],
     [3, 3]], columns=['Expected', 'Predicted'])

print(results)
print()
print(classification_report(results['Expected'], results['Predicted']))
```

To get the following output

```
   Expected  Predicted
0         1          1
1         1          2
2         1          3
3         2          1
4         2          2
5         2          3
6         3          1
7         3          2
8         3          3

             precision    recall  f1-score   support

          1       0.33      0.33      0.33         3
          2       0.33      0.33      0.33         3
          3       0.33      0.33      0.33         3

avg / total       0.33      0.33      0.33         9
```

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answered Nov 23 '18 at 9:08

Steztric

109 2